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Reaction Wheel–Control Surface Composite Attitude Control for Maneuvering Flight in Rarefied Atmosphere

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KEYWORDS

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composite attitude control;
chained-increment torque allocation;
angular-momentum desaturation

Abstract

In near-space rarefied atmosphere, the aerodynamic moment of control surfaces degrades severely, resulting in insufficient attitude-control authority. Reaction wheels provide propellant-free high-precision attitude regulation via angular-momentum exchange, yet their momentum capacity is limited and prone to saturation. To address this contradiction, this paper proposes a composite attitude-control architecture integrating reaction wheels and aerodynamic control surfaces, and develops a cooperative control method. The mechanism is: when aerodynamic effectiveness is insufficient, the wheels supplement real-time attitude control torque; when wheels approach saturation, the control surfaces are opportunistically scheduled—under adequate aerodynamic conditions—to actively desaturate the wheels. A model predictive control algorithm computes the desired total torque from the planned trajectory. A three-layer architecture (decision, allocation, execution) is formulated, together with a chained-increment torque-allocation algorithm and a desaturation strategy. A wheel-based pitch-channel control law is further designed within the nonlinear flight-dynamics framework. Simulation results demonstrate improved maneuverability, reduced settling time, and 0.01° accuracy. This study offers a new solution for high-precision attitude control and rapid maneuvering in rarefied atmosphere.

1. Introduction

Near space (20–100 km), as the transitional region between aviation and astronautics, features extreme rarefaction and complex thermal environment, and therefore constitutes a critical flight envelope for mission execution [1–3]. At these altitudes, atmospheric density decreases to approximately one thousandth of sea-level conditions, which severely degrades the aerodynamic control efficiency of aerodynamic surfaces. Such performance deterioration not only limits attitude maneuverability under aggressive maneuvers, but may also induce response lag, control instability, or even system failure [4–6]. Under high-dynamics mission profiles (e.g., high-speed skip trajectories or long-endurance cruise), sole aerodynamic control becomes insufficient to guarantee fast response and high reliability for near-space attitude control. To overcome the aerodynamic bottleneck, various alternative attitude actuation technologies—such as reaction control systems (RCS), magnetic torquers, and ion thrusters—have been investigated worldwide.

RCS generates control torque by high-velocity propellant ejec-

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tion[7], and can operate effectively in vacuum or rarefied environment. Its millisecond-level response is advantageous for suppressing fast disturbances[8]. However, propellant consumption grows proportionally with mission duration[9], which significantly reduces payload mass fraction. Moreover, due to the pulse-width modulation limit (typically 5–10 ms), the minimum torque resolution hardly exceeds the $10^{-2}Nm$ level, constraining high-precision attitude control.

Magnetic torquers generate magnetic dipole through coil current modulation and interact with the geomagnetic field to produce control torque[10,11]. Their zero-propellant consumption and solid-state structure offer excellent reliability for long-duration missions[12,13]. Nevertheless, controllable torque scales with geomagnetic field strength. In near-space flight (20–100 km), the local geomagnetic field drops to merely 1/8–1/10 of LEO level, causing severe loss of effective torque.

Ion thrusters accelerate ionized inert gas (e.g. Xe) to generate micro-thrust[14–16], with specific impulse about one order of magnitude higher than chemical propulsion[17,18], and without combustion-induced mechanical vibration. Yet the thrust density is typically ¡ 0.1 N/kW, and the multistage ionization/acceleration process induces 300–500 ms of system delay, making them poorly suited for real-time disturbance rejection.

Although RCS, magnetic torquers, and ion thrusters each provide unique advantages, their respective propellant dependence, environmental sensitivity, or energy-conversion latency impose intrinsic constraints under high-dynamics, long-duration compound missions. In contrast, reaction wheels exhibit superior engineering compatibility. By exchanging angular momentum through a high-speed rotor, momentum exchange is internalized within the spacecraft, free of propellant storage, geomagnetic coupling, or plasma generation[19]. In vacuum environments, reaction wheels achieve extremely high pointing accuracy with sub-10-ms current-to-torque actuation latency. Moreover, with multi-wheel topologies and momentum-management algorithms, propellant-free long-lifespan operation is theoretically feasible. However, momentum accumulation and bearing wear at high speed still limit the capability of wheels to independently sustain complex control tasks. This motivates research into dynamic desaturation and composite attitude-control architectures to push beyond the conventional use envelope of inertia-based actuators.

To reconcile the contradiction between the propellant-free nature and the momentum-saturation tendency of reaction wheels, this paper proposes a composite attitude-control strategy for reaction wheel–control surface cooperation in rarefied-atmosphere maneuvering flight. The core idea is dynamic torque allocation and real-time complementary actuation: control surfaces are prioritized for large-amplitude actuation, whereas reaction wheels compensate when aerodynamic efficiency is insufficient; meanwhile, control surfaces opportunistically unload the accumulated angular momentum to prevent saturation and extend in-service duration. This scheme improves control accuracy while balancing energy consumption and redundancy, providing an efficient and reliable control paradigm for near-space maneuvering missions.

The remainder of this paper is organized as follows. Section 2 describes the problem formulation. Section 3 presents the composite wheel–surface control architecture. Section 4 develops the corresponding composite control algorithms. Section 5 reports simulation results that validate the proposed strategy. Section 6 concludes the paper and outlines future research.

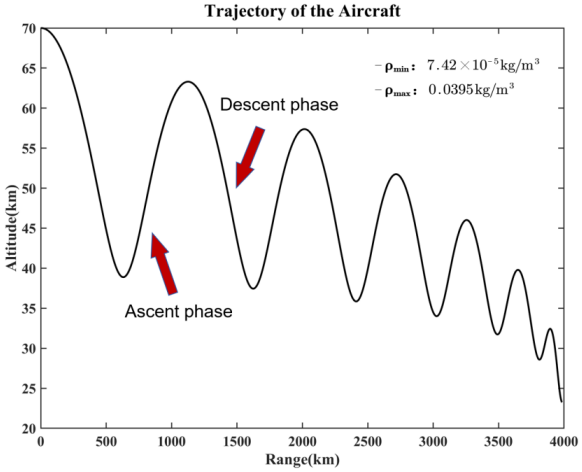


Fig. 1 Variation of vehicle pitch angle during near-space maneuvering skip flight

2. Problem Description

This work focuses on the attitude control problem of a vehicle executing high-maneuver skip trajectories in near space. Such missions typically involve repeated traversing across the atmospheric boundary to achieve flight modes of improved energy efficiency. Represented by the Hypersonic Technology Vehicle 2 (HTV-2) developed by the Defense Advanced Research Projects Agency (DARPA)[20], this class of vehicles adopts a "skip-glide" flight mode, in which multiple upward skips and downward dives are performed after reentry to extend the glide range and maintain hypersonic flight[21]. Fig. 1 illustrates the variation of altitude with respect to ground range in a typical skip trajectory, where the vehicle repeatedly passes through multiple high-altitude segments, forming a complex pattern of successive ascents and descents. During the entire skip-glide process, the atmospheric density experienced by the vehicle varies drastically: when located at lower-altitude segments, the density may reach a maximum of $\rho_{\max} = 0.0395$ kg/m^3, while when the vehicle ascends into the upper atmosphere or even approaches the near-exoatmospheric boundary, the density reduces to a minimum of $\rho_{\min} = 7.42 \times 10^{-5}$ kg/m^3.

During skip-type flight, the vehicle must execute frequent and aggressive attitude maneuvers to match rapid trajectory variations. In particular, sharp pitch-angle adjustments are required during the ascent and descent segments to maintain flight stability and control precision. Fig. 2 shows the representative pitch-angle profile, where the range of variation is large and the rate is high, imposing strict response-performance requirements on the attitude-control system.

However, the near-space environment poses severe challenges to traditional aerodynamic-surface-based control. In high-altitude rarefied air, the available dynamic pressure drops markedly, leading to drastic attenuation of the aerodynamic moment generated by control surfaces[22]. This degradation constrains torque amplitude, narrows control bandwidth, and slows response rate. When the available aerodynamic authority cannot meet mission requirements, response lag, instability, or even mission failure may occur. Fig. 3 compares the desired control torque M_{target} with the aerodynamic moment limits $M_{rd,max}$ during skip maneuvering, showing evident torque-deficit intervals where traditional control mechanisms cannot satisfy maneuverability demand.

Therefore, actuator types independent of aerodynamic conditions, together with composite attitude-control strategies, are required to enhance control capability in low-aerodynamic-efficiency

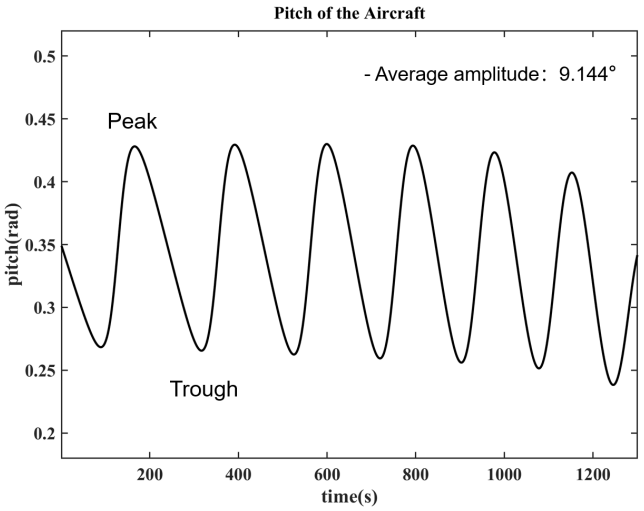


Fig. 2 Variation of vehicle altitude with ground range during near-space maneuvering skip flight

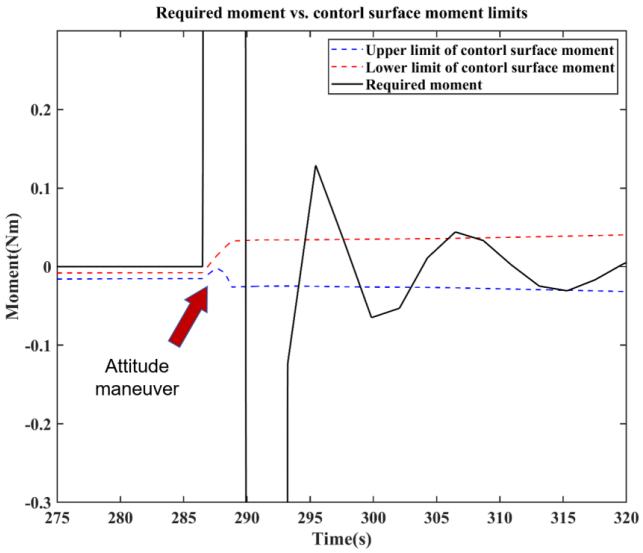


Fig. 3 Comparison of demanded attitude control torque and actuator aerodynamic torque limit during skip-maneuver flight

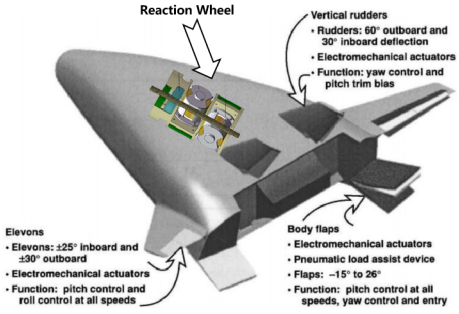


Fig. 4 Schematic of the reaction-wheel / control-surface composite attitude-control actuation architecture

environments. For this reason, this work introduces a reaction-wheel–control-surface composite actuator architecture.

The structural schematic of the composite wheel–surface actuation mechanism is shown in Fig. 4 (side view). The control surface generates aerodynamic moment via aerodynamic deflection and is the primary moment source in conventional atmospheric flight. The reaction wheel (RW), integrated within the vehicle, produces reaction torque through rotor acceleration/deceleration, independent of aerodynamic conditions[23]. Surface deflection adjusts the aerodynamic moment coefficient, whereas the reaction wheel adjusts its rotor angular momentum to produce inertial torque. Although their physical torque-generation mechanisms differ completely, both act on the rotational dynamics channels and can be uniformly combined, scheduled, and allocated in the control law.

In low- and mid-altitude flight, control surfaces—with high aerodynamic efficiency—provide the dominant attitude-control torque. When ascending into near-space rarefied atmosphere where surfaces can no longer sustain sufficient control bandwidth, reaction wheels provide independent inertial torque to compensate for aerodynamic degradation, preventing control loss. Meanwhile, the composite actuator allows residual aerodynamic torque to be used for reaction-wheel desaturation, fundamentally alleviating accumulated momentum saturation and enabling recoverable, sustainable resource scheduling. Compared with a single actuator type, the composite wheel–surface scheme not only broadens torque availability, but also increases controllability and bandwidth across altitude segments, significantly improving attitude-control reliability for cross-altitude near-space maneuvering missions.

3. Composite Attitude-Control Architecture Based on Reaction Wheel and Control Surface

To fully exploit the performance of the RW–CS composite attitude-control actuators, a unified, coordinated and efficient composite control framework is required. As illustrated in Fig. 5, the entire architecture is divided into three functional layers: Decision Layer, Allocation Layer, and Execution Layer. The Decision Layer determines the composite control mode (Composite Control Mode, CCM) based on current environment parameters, actuator states, and mission requirements. Meanwhile, it generates the demanded attitude control torque M_{target} subject to both attitude constraints and actuator physical limitations. Under different CCMs, the Allocation Layer distributes M_{target} to the reaction wheels and control surfaces, converting it into demanded control-surface deflection δ_{rd} and reaction-wheel torque command M_{rw}. Finally, the Execution Layer implements the closed-loop control of δ_{rd} and M_{rw}, driving actuators to track the assigned control commands.

3.1. Decision Layer

In the composite attitude–control framework, the decision layer functions as the "command nucleus" of the system, responsible for global reasoning and generation of control directives. Its primary objective is to select the optimal control mode across varying flight environments based on the vehicle state, actuator capability, and mission requirements, and to generate in real time the total required torque that satisfies the attitude–control objective.

Within the operation of the decision layer, the input information behaves like a multidimensional sensor network that encapsulates the key state variables of the vehicle. The flow of this information provides the basis for mode selection and decision making, as summarized in Table 1.

Depending on the environment parameters and actuator states, the Decision Layer outputs different composite attitude-control modes.

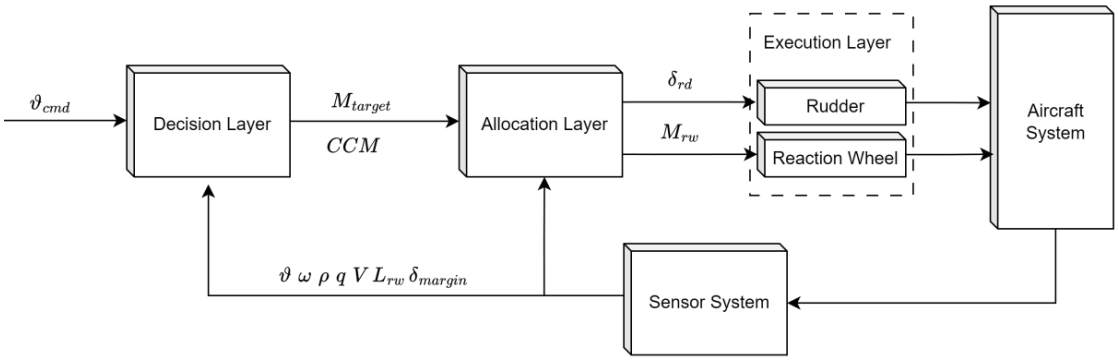


Fig. 5 Architecture of the reaction-wheel/control-surface composite attitude-control framework

Table 1 Input information of the Decision Layer

Input	Description
Mission command	Desired attitude angle θ_{cmd} used to generate reference torque M_{target}
Attitude state	Current attitude angle θ and angular rate ω
Environment parameters	Altitude H, atmospheric density ρ, dynamic pressure q
Actuator state	Control-surface torque margin δ_{margin}, RW angular momentum L_{rw}

Table 2 Output composite control modes of the Decision Layer

Cond. 1	Cond. 2	Output mode	Mode ID
$\delta_{\text{margin}} > 0$	RW unsaturated	Surface-only mode	CCM1
$\delta_{\text{margin}} > 0$	RW saturated	Surface-dominant RW-unloading mode	CCM2
$\delta_{\text{margin}} \le 0$	RW unsaturated	Composite RW–surface mode	CCM3
$\delta_{\text{margin}} \le 0$	RW saturated	Emergency mode	CCM4

When the vehicle is within a dense-atmosphere regime such that sufficient aerodynamic torque margin exists ($\delta_{\text{margin}} > 0$), and the reaction wheel is not saturated, the controller selects CCM1 (surface-only control). In this case, attitude control is provided exclusively by the aerodynamic control surface, while the reaction wheel does not contribute to torque generation.

When $\delta_{\text{margin}} > 0$ while the reaction wheel momentum is saturated, the system switches to CCM2 (surface-dominant with RW-unloading). In this mode, the aerodynamic surface provides attitude control while simultaneously generating a reverse aerodynamic moment to unload the accumulated RW angular momentum.

When the flight altitude increases and the atmosphere becomes thin, resulting in degraded aerodynamic effectiveness such that $\delta_{\text{margin}} \le 0$, CCM3 (RW–surface composite control) is activated. Both actuators jointly generate control torque to satisfy the attitude-control requirement.

In extreme emergency conditions where both actuators reach their saturation limits — i.e. neither the aerodynamic surface nor the reaction wheel can further provide effective incremental torque — CCM4 is triggered. This mode forces both actuators into saturation protection, ensuring that the control system transitions into a safe fail-protection state while preventing irreversible hardware overload.

After the control mode is determined, the Decision Layer generates the total demanded attitude-control torque according to mission requirements and current flight states. Based on the attitude kinematics, the controller solves the following control objective:

$$M = J\dot{\omega} + \omega \times (J\omega) \tag{1}$$

where J denotes the inertia matrix of the vehicle, ω is the angular velocity, and M is the external torque acting on the vehicle. This process must not only ensure the reachability of the attitude command, but also embed various hardware physical constraints, including actuator maximum torque / surface deflection limits, servo and reaction–wheel motor bandwidth (rate limits), as well as reaction–wheel angular–momentum saturation boundaries. These constraints are explicitly incorporated into the MPC prediction model in the form of input constraints or augmented states, such that the obtained optimal solution remains physically executable.

To achieve globally optimal control, the Decision Layer employs Model Predictive Control (MPC)[24] as the optimization solver. By minimizing the predicted tracking error and the control–energy consumption, MPC computes the optimal torque input sequence in a rolling horizon manner.

$$\begin{aligned} \min \quad & \sum_{k=0}^{N-1} \|x^{(k)} - x_{\text{ref}}^{(k)}\|_Q^2 + \|u^{(k)}\|_R^2 + \|x^{(N)} - x_{\text{ref}}^{(N)}\|_F^2 \\ \text{s.t.} \quad & x_{\min} \le x^{(k)} \le x_{\max}, \quad k = 0, 1, \ldots, N, \\ & u_{\min} \le u^{(k)} \le u_{\max}, \quad k = 0, 1, \ldots, N-1. \end{aligned} \tag{2}$$

where $\mathbf{x}^{(k)}$ denotes the predicted state at step k, $u^{(k)}$ is the control input, Q and R are the weighting matrices for state error and control effort, and F is the terminal state error weighting matrix. $\mathbf{x}_{\min}$, $\mathbf{x}_{\max}$, $\mathbf{u}_{\min}$ and $\mathbf{u}_{\max}$ define the lower and upper bounds of the system states and control inputs, respectively. N is the prediction horizon length. Solving (2) yields the optimal input sequence U^*, and its first element is taken as the demanded attitude–control torque:

$$M_{\text{target}} = U^*(0) \tag{3}$$

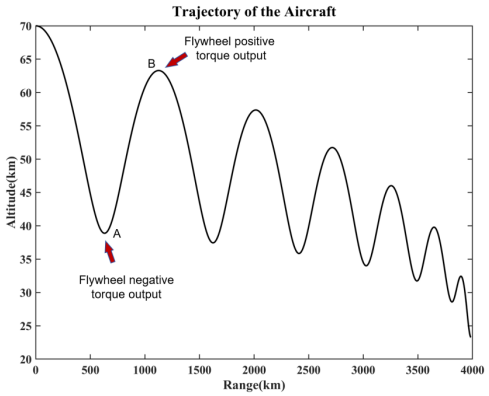


Fig. 6 Schematic of reaction–wheel angular–momentum pre–bias mechanism

The introduction of MPC enables the Decision Layer not only to consider the current attitude error, but also to foresee the future control capability and system dynamics, thus generating predictive control commands.

Based on this feature, a reaction–wheel angular–momentum pre–bias mechanism is designed to dynamically adjust the RW momentum and maximize its actuation capability.

Let the optimal control sequence obtained by MPC be $\mathbf{U}^* = [\,\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{N-1}\,]^T$. If the maximum aerodynamic surface torque available at step k, $M_{\mathrm{rd}}^{(k)}$, is smaller than the demanded control torque command u_k, while the aerodynamic margin is still sufficient, then the system forces the RW to output a negative torque — converting the surplus aerodynamic torque into stored RW momentum for future use (i.e., momentum pre–bias).

The concept of RW momentum pre–bias and release is illustrated in Fig. 6. At the valley region "A", where the control surface has surplus torque capability, the RW applies a negative torque to store this surplus as angular momentum. When the vehicle reaches the peak region "B", this stored momentum can then be released to generate more positive torque.

Through this strategy, the RW momentum is no longer passively accumulated; instead, it becomes an actively managed, schedulable energy buffer, significantly improving subsequent control capability and resource allocation flexibility.

3.2. Allocation Layer

The Allocation Layer serves as a bridge between the upper–level task commands and the lower–level actuators. Its primary goal is to distribute the demanded attitude–control torque to heterogeneous actuators (e.g., aerodynamic surfaces and reaction wheels) in a rational and efficient manner, so that system redundancy can be fully utilized and both robustness and accuracy can be ensured.

According to the Composite Control Mode (CCM) determined in the Decision Layer, the Allocation Layer executes different torque–assignment strategies, as summarized in Table 3. The CCMs correspond to those defined in Table 2.

In Table 3, M_{target} is the demanded attitude–control torque, M_{rd} and M_{rw} denote the torque assigned to the aerodynamic surface and reaction wheel, respectively, M_{unload} is the active RW unloading torque, and $M_{\mathrm{rd,max}}, M_{\mathrm{rw,max}}$ denote their respective maximum torque limits.

After torque allocation, the torque commands must be converted into actuator–specific control inputs, consistent with their physical actuation channels.

Table 3 Dynamic torque allocation strategies under different CCMs

CCM	Dynamic Allocation Strategy
CCM1	Assign the demanded attitude–control torque entirely to the aerodynamic surface: $M_{\mathrm{rd}} = M_{\mathrm{target}}$
CCM2	Set the RW unloading torque as $M_{\mathrm{rw}} = -M_{\mathrm{unload}}$, and compensate via surface: $M_{\mathrm{rd}} = M_{\mathrm{target}} + M_{\mathrm{unload}}$
CCM3	Use the chained–increment criterion: $M_{\mathrm{target}} = M_{\mathrm{rd}} + M_{\mathrm{rw}}$
CCM4	Drive both actuators to saturation: $\|M_{\mathrm{rd}}\| = M_{\mathrm{rd,max}}$, $\|M_{\mathrm{rw}}\| = M_{\mathrm{rw,max}}$

(1) RW control channel

Reaction wheels operate in torque–tracking mode. Thus, the assigned M_{rw} can be directly fed into the RW controller without additional processing.

(2) Aerodynamic surface control channel

The control input of control surfaces is the deflection angle δ. Therefore, the surface torque M_{rd} must be mapped back to the corresponding deflection.

The aerodynamic moment is expressed as

$$M = \frac{1}{2}\rho V^2 S\bar{c}\, C_m(\delta, \alpha, M_a, h, q) \tag{4}$$

where ρ is air density; V is freestream velocity; S is reference area; $\bar{c}$ is mean aerodynamic chord; and C_m is the pitching–moment coefficient, dependent on α, Mach number M_a, altitude h, body–rate q and deflection δ.

Given the allocated torque, the pitching–moment coefficient can be back–computed as

$$C_m = \frac{2M}{\rho V^2 S\bar{c}} \tag{5}$$

Based on flight–state parameters (measured in real time), the system then retrieves the corresponding deflection angle via the inverse aerodynamic map:

$$\delta = \mathcal{F}_{\mathrm{inv}}(C_m, \alpha, M_a, h, q) \tag{6}$$

where $\mathcal{F}_{\mathrm{inv}}(\cdot)$ denotes the look–up/interpolation based inverse characteristic of the control surface.

3.3. Execution Layer

Within the composite attitude-control architecture, the Execution Layer is the key stage that physically implements the control commands and ensures accurate attitude tracking. This layer receives the commanded surface deflection angles or torque commands from the Allocation Layer, and drives the actuators (control surface and reaction wheel) in a fine–closed–loop manner to adjust the vehicle attitude in real time.

(1) Control Surface Regulation

The role of the aerodynamic control surface in the Execution Layer is to generate attitude-control torque by adjusting the surface deflection angle. The Execution Layer receives the demanded surface deflection generated by the Allocation Layer, and regulates the surface position through a closed–loop actuator servo (e.g., PID-based surface servo) to precisely track the commanded attitude profile[25].

(2) Reaction–Wheel Regulation

The reaction wheel provides additional attitude-control authority by modulating its angular momentum, especially in flight regimes where the aerodynamic control surface becomes insufficient or when faster transient torque is required. The Execution Layer receives the demanded reaction–wheel torque from the Allocation Layer, and regulates the wheel torque/velocity through a closed–loop motor control loop in real time[26].

4. MPC-CI Based Reaction Wheel–Control Surface Composite Attitude Control Torque Computation and Allocation Algorithm

To systematically implement the concept of reaction wheel (RW)–control surface (CS) composite attitude control, Chapter 3 established a hierarchical composite attitude control framework, dividing the control system into three layers: Decision Layer, Allocation Layer, and Execution Layer. This framework effectively integrates mission requirements, actuator performance, and flight state information, offering good scalability and adaptability.

Based on this framework, this chapter further develops control algorithms corresponding to different composite control modes. Specifically, it discusses the MPC formulation and solution in the Decision Layer, the chain-incremental allocation method in the Allocation Layer, and the RW/CS control algorithms in the Execution Layer, aiming to achieve a predictive, adaptive, and efficiently coordinated control system capable of supporting attitude stability and rapid response under thin-atmosphere maneuvering flight.

4.1. MPC-Based Desired Attitude Control Torque Computation

In this study, only the pitch channel is considered, while lateral channels are set to zero to avoid roll-pitch-yaw coupling that may obscure algorithm and actuator scheduling mechanisms. During the hopping-gliding task, the pitch channel dominates aerodynamic lift-drag regulation and energy management; its dynamics are significantly stronger than roll and yaw, and the vehicle trajectory is approximately symmetric, so lateral disturbances can be neglected. Under this simplification, the attitude dynamics reduce to a single-pitch channel model:

$$\begin{cases} \dot{\vartheta} = \omega_z \\ \dot{\omega}_z = \frac{u}{I_z} \end{cases} \tag{7}$$

where ϑ is the pitch angle, ω_z is the angular rate around the z-axis, and I_z is the moment of inertia about the z-axis. Define

$$x_1 = \vartheta, \quad x_2 = \omega_z \tag{8}$$

The state-space model can then be expressed as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{I_z} \end{bmatrix} u \tag{9}$$

The Decision Layer computes the total required torque for tracking commands without considering the nonlinear mapping between torque and control surface, reducing the model to a linear time-invariant system. Using zero-order hold (ZOH)[27], the system is discretized with sampling period T:

$$x_{k+1} = A_d x_k + B_d u_k \tag{10}$$

where the discrete-time matrices A_d and B_d are obtained via:

$$\begin{bmatrix} A_d & B_d \\ 0 & I \end{bmatrix} = \exp\left(\begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix} T \right) \tag{11}$$

The quadratic cost function for MPC is defined as:

$$\begin{aligned} J = \sum_{i=0}^{N-1} \Big[(x_i - x_{\mathrm{ref},i})^{\mathrm{T}} Q (x_i - x_{\mathrm{ref},i}) + u_i^{\mathrm{T}} R u_i \Big] \\ + (x_N - x_{\mathrm{ref},N})^{\mathrm{T}} F (x_N - x_{\mathrm{ref},N}) \end{aligned} \tag{12}$$

where N is the prediction horizon. $x_{\mathrm{ref},i}$ denotes the reference state trajectory; for the first state x_1, its reference trajectory corresponds to the commanded pitch angle ϑ_{cmd}. Q is the state–error weighting matrix, ensuring reference–tracking performance; R is the control–input weighting matrix, suppressing excessive control effort; and F is the terminal state–error weighting matrix, ensuring that the predicted terminal state lies within the region of attraction.

In practice, Q, R and F are typically chosen as diagonal matrices, e.g.,

$$Q = \mathrm{diag}(q_1, q_2)$$

where q_1 corresponds to the pitch–angle error, and q_2 corresponds to the pitch–rate error. To guarantee closed–loop convergence, the terminal–state weighting matrix F should be selected with a sufficiently large magnitude, which is typically chosen as a scaled version of Q, i.e. $F = nQ$, where n usually ranges from 10 to 100.

Based on the discretized model (10), the system states can then be propagated over the prediction horizon as

$$\begin{aligned} x_1 &= A_d x_0 + B_d u_0 \\ x_2 &= A_d x_1 + B_d u_1 = A_d^2 x_0 + A_d B_d u_0 + B_d u_1 \\ &\vdots \\ x_N &= A_d^N x_0 + \sum_{i=0}^{N-1} A_d^{N-1-i} B_d u_i \end{aligned}$$

$\mathbf{x}_0$ denotes the current system state vector, obtained from sensor measurements. Define the state–transition matrix Φ and the control–influence matrix Ψ as

$$\Phi = \begin{bmatrix} 1 \\ A_d \\ A_d^2 \\ \vdots \\ A_d^N \end{bmatrix}, \quad \Psi = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ B_d & 0 & \cdots & 0 \\ A_d B_d & B_d & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A_d^{N-1} B_d & A_d^{N-2} B_d & \cdots & B_d \end{bmatrix} \tag{13}$$

Then the state sequence can be written as:

$$X = \Phi x_0 + \Psi U \tag{14}$$

where the stacked state sequence and control sequence are defined as $\mathbf{X} = [x_0^{\mathrm{T}}, x_1^{\mathrm{T}}, x_2^{\mathrm{T}}, \ldots, x_N^{\mathrm{T}}]^{\mathrm{T}}$ and $\mathbf{U} = [u_0^{\mathrm{T}}, u_1^{\mathrm{T}}, \ldots, u_{N-1}^{\mathrm{T}}]^{\mathrm{T}}$, respectively. The original quadratic cost function (12) can then be rewritten as:

$$J = (X - X_{\mathrm{ref}})^{\mathrm{T}} \bar{Q} (X - X_{\mathrm{ref}}) + U^{\mathrm{T}} \bar{R} U \tag{15}$$

with

$$\begin{aligned} &\bar{\mathbf{Q}} = \mathrm{diag}(\mathbf{Q}, \mathbf{Q}, \ldots, \mathbf{Q}, \mathbf{F}) \in \mathbb{R}^{(N+1)n_x \times (N+1)n_x}, \\ &\bar{\mathbf{R}} = \mathrm{diag}(\mathbf{R}, \mathbf{R}, \ldots, \mathbf{R}) \in \mathbb{R}^{N n_u \times N n_u}, \\ &\mathbf{X}_{\mathrm{ref}} = [\, x_{\mathrm{ref},0}^{\mathrm{T}}, x_{\mathrm{ref},1}^{\mathrm{T}}, x_{\mathrm{ref},2}^{\mathrm{T}}, \ldots, x_{\mathrm{ref},N}^{\mathrm{T}} \,]^{\mathrm{T}}. \end{aligned}$$

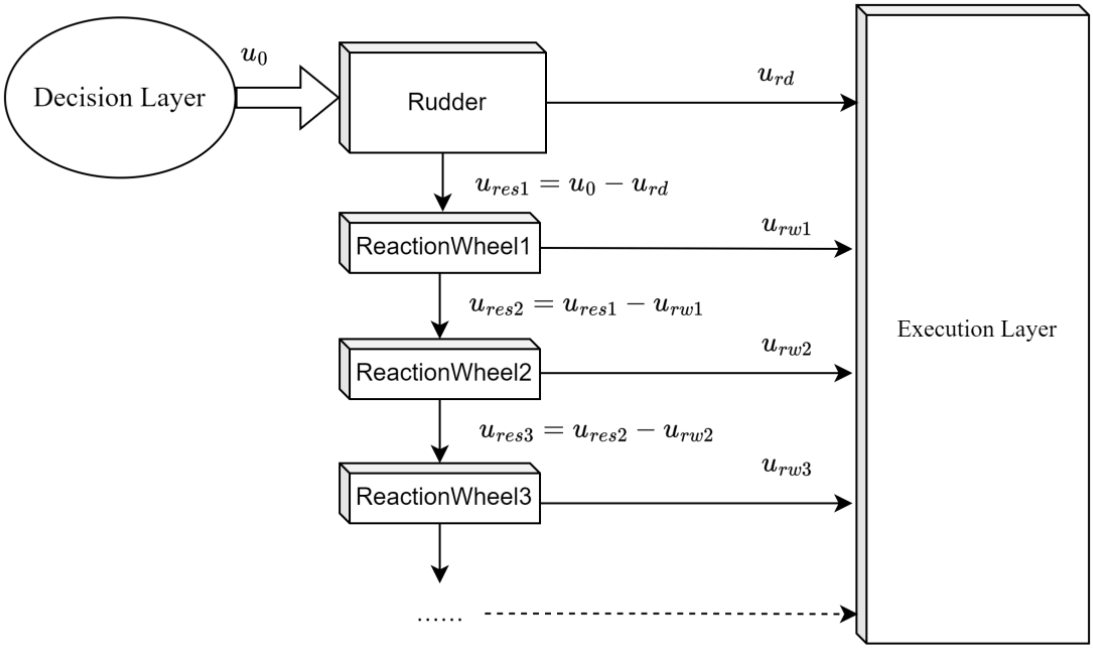


Fig. 7 Illustration of dynamic torque allocation based on chained-incremental strategy.l framework

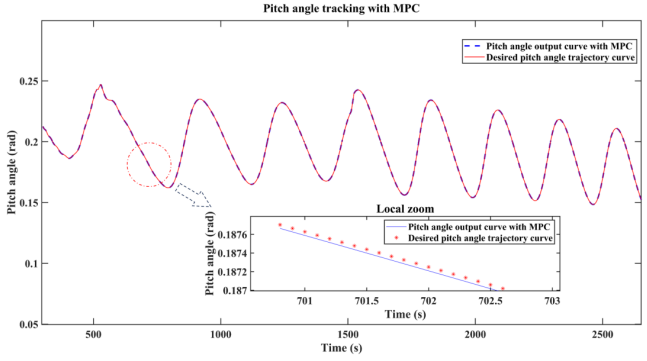


Fig. 8 MPC-based pitch angle tracking in the Decision Layer

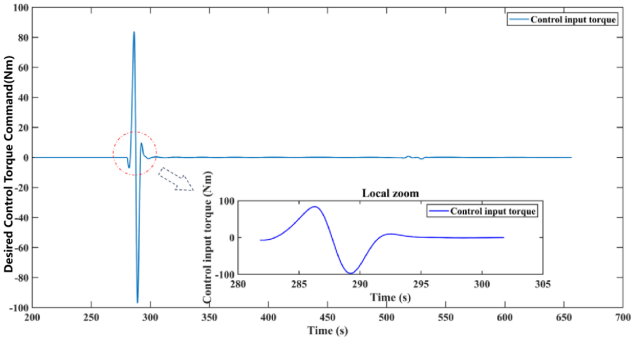


Fig. 9 MPC-based desired control torque in the Decision Layer

Substituting the state prediction model (14) into (15), we have

$$\mathbf{J} = (\Phi\mathbf{x}_0 + \Psi\mathbf{U} - \mathbf{X}_{\text{ref}})^{\mathrm{T}}\bar{\mathbf{Q}}(\Phi\mathbf{x}_0 + \Psi\mathbf{U} - \mathbf{X}_{\text{ref}}) + \mathbf{U}^{\mathrm{T}}\bar{\mathbf{R}}\mathbf{U}. \tag{16}$$

Expanding and rearranging yields

$$\begin{aligned}\mathbf{J} = {} & \mathbf{U}^{\mathrm{T}}(\Psi^{\mathrm{T}}\bar{\mathbf{Q}}\Psi + \bar{\mathbf{R}})\mathbf{U} + 2(\Psi^{\mathrm{T}}\bar{\mathbf{Q}}(\Phi\mathbf{x}_0 - \mathbf{X}_{\text{ref}}))^{\mathrm{T}}\mathbf{U} \\ & + (\Phi\mathbf{x}_0 - \mathbf{X}_{\text{ref}})^{\mathrm{T}}\bar{\mathbf{Q}}(\Phi\mathbf{x}_0 - \mathbf{X}_{\text{ref}})\end{aligned} \tag{17}$$

Define the quadratic coefficient matrix

$$\mathbf{H} = \Psi^{\mathrm{T}}\bar{\mathbf{Q}}\Psi + \bar{\mathbf{R}},$$

and the linear vector

$$\mathbf{f} = \Psi^{\mathrm{T}}\bar{\mathbf{Q}}(\Phi\mathbf{x}_0 - \mathbf{X}_{\text{ref}}).$$

Since the third term in (17) is constant, the optimization problem can be converted into a standard quadratic programming (QP) form:

$$\min_{\mathbf{U}} \frac{1}{2}\mathbf{U}^{\mathrm{T}}\mathbf{H}\mathbf{U} + \mathbf{f}^{\mathrm{T}}\mathbf{U}, \tag{18}$$

where $\mathbf{U} = [u_0, u_1, \ldots, u_{N-1}]^{\mathrm{T}}$. To improve the practical feasibility and robustness of the model predictive controller, constraints on both the control inputs and system states must be explicitly included in the optimization. Here, the control input u_k represents the total pitch control torque, which is generated by the combination of aerodynamic surface torque and reaction wheel torque, both subject to physical limitations.

(1) Control Input Constraints

The control input limits are determined jointly by the available aerodynamic surface torque and the maximum/minimum output torque of the reaction wheel under the current flight state, i.e.,

$$u_{\min} \le u_k \le u_{\max}.$$

Specifically, the constraints are constructed as follows:

- **Aerodynamic surface torque range:** Using the control surface deflection limits ($\delta_{\max}$, $\delta_{\min}$) and the aerodynamic model, the corresponding minimum and maximum available torques are obtained as $M_{\text{aero,min}}$ and $M_{\text{aero,max}}$.

Table 4 Steps for dynamic torque allocation based on chained-incremental method

Step	Description	Maximum Torque Output	Residual Torque Transfer
Allocate to surface	Assign as much control torque as possible to the aerodynamic surface until its maximum output is reached.	$\mathbf{M}_{\mathrm{rd,max}}$	If the surface is saturated, the remaining torque is transferred to Reaction Wheel 1.
Allocate to RW1	Assign the remaining torque to Reaction Wheel 1, computing its maximum torque $M_{\mathrm{rw1,max}}$. If RW1 cannot provide sufficient torque, the remainder is passed to RW2.	$M_{\mathrm{rw1,max}}$	If RW1 saturates, residual torque transfers to RW2.
Allocate to RW2	Similarly, assign torque to RW2 following the same logic; any remaining torque is passed to RW3 until RW2 reaches its maximum output.	$M_{\mathrm{rw2,max}}$	If RW2 saturates, residual torque transfers to RW3.
Allocate to RW3	If RW3 still cannot provide enough torque, the control system adjusts according to the designed strategy.	$M_{\mathrm{rw3,max}}$	If RW3 saturates, remaining torque is handled by other actuators according to the system strategy.

- **Reaction wheel torque range:** Considering angular momentum saturation, the reaction wheel output torque is constrained not only by motor capacity but also by the current momentum state h and the saturation limit $h_{\max}$. If the reaction wheel torque is denoted as M_{rw}, then

$$M_{\mathrm{rw,max}} = \mu \frac{h_{\max} - h}{\Delta t}, \quad M_{\mathrm{rw,min}} = \mu \frac{-h_{\max} - h}{\Delta t},$$

  where $\mu \in (0, 1]$ is a safety factor to prevent irreversible saturation, and Δt is the control period.

- **Total control input limits:** The overall control input constraints are given by

$$u_{\max} = M_{\mathrm{aero,max}} + M_{\mathrm{rw,max}}, \quad u_{\min} = M_{\mathrm{aero,min}} + M_{\mathrm{rw,min}}. \tag{19}$$

The above limits are updated in real time during each MPC optimization according to the current flight state (reaction wheel angular momentum and aerodynamic surface deflection limits).

(2) Pitch Angle State Constraints

To avoid stall conditions, the pitch angle ϑ is limited. Considering the linear approximation between the angle of attack α and pitch angle ϑ:

$$\alpha = \vartheta - \theta,$$

where θ is the flight path angle. Therefore, the pitch angle constraints can be derived from the angle-of-attack limits $\alpha_{\min}$ and $\alpha_{\max}$:

$$\vartheta_{\min} = \alpha_{\min} + \theta, \quad \vartheta_{\max} = \alpha_{\max} + \theta,$$

and can be expressed as

$$\vartheta_{\min} \le x_{1,k} \le \vartheta_{\max}, \quad \forall k \in [0, N]. \tag{20}$$

The resulting optimization problem is written as

$$\begin{aligned} \min_{\mathbf{U}} \quad & J = \mathbf{U}^{\mathrm{T}}\mathbf{H}\mathbf{U} + 2\mathbf{f}^{\mathrm{T}}\mathbf{U}, \\ \text{s.t.} \quad & u_{\min} \le u_k \le u_{\max}, \quad k = 0, \ldots, N-1, \\ & \vartheta_{\min} \le x_{1,k} \le \vartheta_{\max}, \quad k = 1, \ldots, N, \\ & x_{k+1} = A_d x_k + B_d u_k, \quad x_0 \text{ known}. \end{aligned} \tag{21}$$

The optimization is solved using the Interior Point Method (IPM)[28], and a receding-horizon strategy[29] is adopted, executing only the current optimal input at each step and then updating the problem for the next iteration. In real-time control, computational efficiency is critical. IPM can converge to high-precision solutions in polynomial time for small to medium-scale QP problems (with tens to hundreds of optimization variables), allowing typical control sampling periods (10–100 ms) to meet real-time requirements.

4.2. Chained-Incremental Torque Allocation in CCM3

In the CCM_3 composite attitude control mode, the desired attitude control torque computed by the Decision Layer is allocated to each actuator using a *chained-incremental* method[30]. This process ensures that control loads are distributed according to actuator priority, allowing the system to gradually engage lower-priority actuators when higher-priority ones reach their limits. Figure 7 illustrates the basic principle of chained-incremental allocation.

Before allocation, the priority of each actuator must be defined. High-priority actuators are assigned control loads first, while lower-priority actuators are engaged only when all preceding actuators have reached saturation. In this work, the priority order is defined as follows:

- **Aerodynamic surface:** highest priority
- **Reaction wheel 1:** second-highest priority
- **Reaction wheel 2:** lower than wheel 1
- **Reaction wheel 3:** lowest priority

The core idea of this strategy is that lower-priority actuators (e.g., reaction wheels) are activated only when high-priority actuators (e.g., aerodynamic surfaces) cannot provide sufficient control torque. Once the priorities are determined, the allocation proceeds according to the steps summarized in Table 4.

5. Simulation Analysis

5.1. Pure Software Simulation

The reaction wheel (RW)–control surface (CS) composite attitude control simulation model is built in MATLAB 2020a. The simulation step size is set to 0.1 s. The RW maximum torque is limited to

60 Nm, the maximum angular momentum does not exceed 240 Nms, and the control surface deflection is restricted within $-30^\circ \sim 30^\circ$.

(1) Decision Layer Desired Torque Simulation

The prediction horizon is set to 40 steps. The cost function weighting matrices are:

$$Q = \begin{bmatrix} 10^4 & 0 \\ 0 & 100 \end{bmatrix}, \quad R = 0.001, \quad F = 10 \begin{bmatrix} 10^4 & 0 \\ 0 & 100 \end{bmatrix}$$

Simulation is run to track the input pitch angle command curve. The results are shown in Fig. 8.

As shown in Fig. 8, the MPC system achieves high-precision tracking of pitch commands under continuously fluctuating test conditions, with steady-state error below 0.01° and phase lag within 50 ms. The robustness comes from the receding horizon optimization: the controller predicts future states over N steps and dynamically optimizes control inputs via the quadratic cost function.

Fig. 9 shows the optimal control torque profile of the composite attitude control mechanism, further revealing the optimization characteristics of the MPC. The torque output exhibits a typical quasi-square wave form, with peak torque occurring during the transient adjustment phase around $t = 280$ s, while automatically converging to a low-energy region below $0.1\,\mathrm{N \cdot m}$ during steady-state tracking. This time-varying gain characteristic indicates that the MPC achieves a Pareto-optimal balance between tracking performance and control cost, where the weighting matrix of the control inputs in the cost function effectively suppresses unnecessary energy consumption.

(2) Chain-Incremental Dynamic Torque Allocation Simulation

The chain-incremental dynamic torque allocation within the MPC prediction horizon is simulated, with prediction steps increased to 60 for clarity. Results are shown in Fig. 10.

In Fig. 10 during the initial phase (prediction steps 0–15, corresponding to actual time 0–1.5 s), the aerodynamic control surfaces generate pitch torque that fully meets the trajectory tracking requirement, while the reaction wheel actuators remain at zero torque output. The system operates in the single-actuator mode CCM1. When the prediction horizon reaches step 16 ($t = 1.6$ s), the aerodynamic surfaces approach their physical saturation limits ($\delta \gtrsim \pm 30^\circ$), so the available torque from the surfaces alone is insufficient to satisfy the tracking demand. According to the predefined chained-increment allocation strategy, the torque deficit is dynamically assigned to the reaction wheels, forming the composite control mode CCM3 (aerodynamic surfaces + reaction wheels).

Fig. 11, showing the variation of control surface deflection, provides a mechanistic verification of Figure 10. During prediction steps 16–55 ($t = 1.6$–5.5 s), the deflection remains at the saturation limit of -30°, providing the maximum aerodynamic authority under rarefied atmospheric conditions. When the aerodynamic environment improves (dynamic pressure increases at $t = 5.5$ s), the surface effectiveness is restored. As seen in the final segment of Figure 10, the reaction wheels cease torque output, confirming the smooth mode switching capability of the allocation algorithm.

(3) RW Angular Momentum Preload Simulation

Fig. 12illustrates the reaction wheel's angular momentum variation and output torque during periods when aerodynamic surface effectiveness is insufficient. As shown, the reaction wheel employs a reverse preload strategy to significantly enhance the system's dynamic performance during attitude maneuvers. In the pre-maneuver phase (segment A), the reaction wheel generates a reverse torque, increasing its angular momentum from 0 to $35.6\,\mathrm{N \cdot m \cdot s}$. During the maneuver phase (segment B), this preloaded angular momentum is released bidirectionally, effectively extending the wheel's operational time and maximizing its contribution to attitude control. This strategy ensures that the reaction wheel can provide larger transient control torques precisely when aerodynamic surfaces alone are inadequate.

(4) RW–CS Composite Attitude Control Simulation

The re-entry phase starts at $t = 155$ s, simulation begins from $t = 102$ s. The composite control system tracks pitch commands, results in Fig. 13.

Analysis of the tracking error curve in Fig. 13 indicates that the composite attitude control system exhibits excellent control precision during the nominal attitude-holding phase. The average tracking error is only 0.0068° (standard deviation $\sigma = 0.0012^\circ$), with 99.7% of the data points distributed within a steady-state error band of $\pm 0.0035^\circ$, fully meeting the high-precision requirement of $< 0.01^\circ$ for Earth observation missions. During the rapid attitude maneuver phase around $t = 280$–290 s, the system experiences aerodynamic disturbance torque spikes, actuator response delays, and inertia parameter perturbations, causing the tracking error to briefly rise to 2.19°. Nevertheless, through the multi-constraint objective function design of the Model Predictive Control (MPC)—which integrates feedforward compensation and covariance-adaptive adjustment mechanisms—the system recovers the error within 5.4 s, with overshoot limited to less than 4.7%.

5.2. Single-Axis HIL Verification of Hybrid Attitude Control and RW Momentum Unloading

To validate the attitude control performance under the reaction–wheel/jet hybrid actuation mechanism, a single-axis high-fidelity hardware-in-the-loop (HIL) platform was constructed. The platform integrates an RW-200S single-axis reaction wheel, a jet actuator composed of an SMC ITV2050-212 proportional pressure regulator and a FeSto MHI9 solenoid valve, a low-friction air-bearing turntable, a gas supply system, and an STM32F407-based real-time controller. Control laws are implemented in MATLAB/Simulink, while data handling uses a hybrid Python–LabVIEW architecture. The RW loop runs at 5 ms, the jet loop at 10 ms, and system sampling at 1 ms, enabling synchronized timing and ensuring hybrid multi-actuator coordination. The final platform is shown in Fig. 14.

The HIL experiment proceeds as follows: an external disturbance torque is first injected via the jet channel to emulate an unsteady background environment. The reaction wheel then performs the attitude tracking command from 0° to 30°. As the angular momentum gradually approaches the wheel's storage capacity, the jet actuator is triggered to unload momentum. Throughout the process, attitude angle, angular velocity, RW torque, and angular momentum evolution are recorded for assessing the dynamic consistency of hybrid actuation during state transition. Typical time-domain responses are shown in Figs. 15–18.

The results show that the RW can still complete the 0°–30° tracking task under jet-induced disturbance, with continuous dynamics and negligible overshoot. Once the stored momentum approaches saturation, jet-based unloading effectively reduces RW angular momentum while maintaining acceptable attitude deviation. After unloading, the attitude rapidly converges without secondary oscillation. These outcomes confirm that, within the same closed-loop controller, the RW and jet actuators can switch actuation modes seamlessly without coupling conflict, while preserving dynamic consistency. Overall, the single-axis HIL platform reliably reproduces hybrid momentum management and attitude stabilization, and provides a transferable physical basis for extending hybrid control strategies to multi-axis spacecraft.

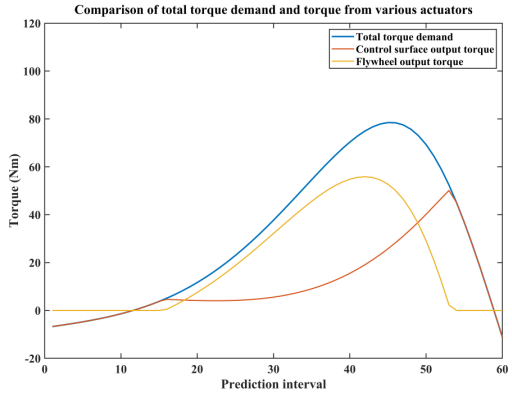


Fig. 10 Chain-incremental dynamic torque allocation

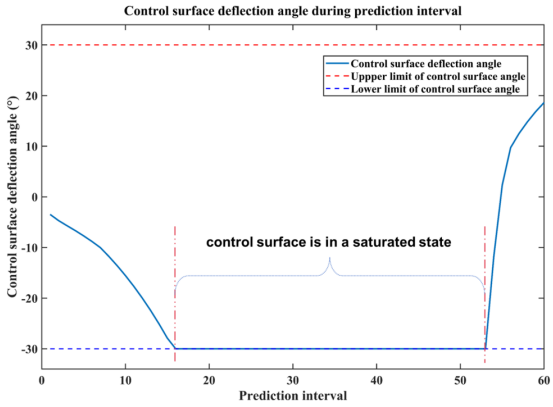


Fig. 11 Control surface deflection under limited aerodynamic authority

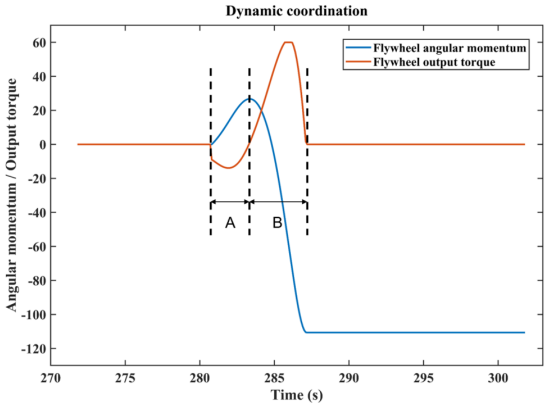


Fig. 12 RW torque and angular momentum under preload strategy

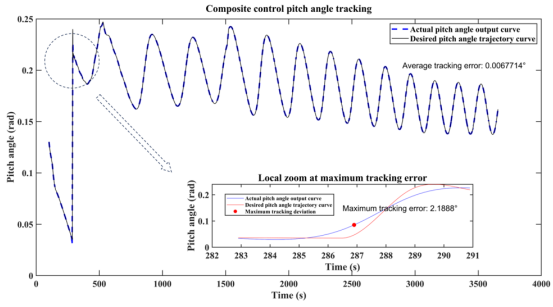


Fig. 13 Pitch angle tracking under RW–CS composite control

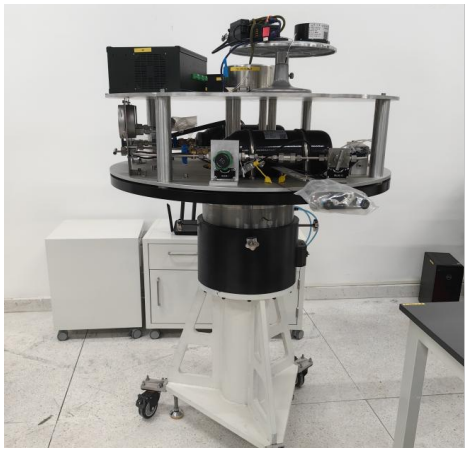


Fig. 14 Single-axis HIL platform for attitude control and RW unloading

6. Conclusions

To address the insufficient control authority of aerodynamic surfaces during maneuvering flight in rarefied atmospheres, this paper proposes a reaction wheel (RW)–control surface (CS) composite attitude control architecture and associated control methodology. The main contributions of this work are summarized as follows:

1. A three-layer RW–CS composite attitude control framework comprising the decision layer, allocation layer, and execution layer is proposed. This architecture eliminates the structural risks inherent in traditional single-actuator control under extreme aerodynamic conditions, extending the control law from a "single-channel / single-actuator response mode" to an integrated "behavior–resource decision mode." When aerodynamic surfaces are insufficient, the RW system can immediately supplement the required attitude control torque to maintain precision. Conversely, when RW angular momentum approaches saturation, the algorithm actively utilizes aerodynamic surfaces to unload momentum during effective aerodynamic phases, restoring RW capacity and closing the resource loop, thereby achieving dynamic complementarity and self-consistency under the dual objectives of attitude control and momentum management.
2. A dynamic allocation algorithm for attitude control torque across the full operating range of aerodynamic surfaces and RWs is developed. Based on a planned trajectory, the decision layer first determines the total attitude torque demand using model predictive control (MPC). Then, the allocation layer dynamically distributes the control load according to a chain-incremental priority criterion, forming collaborative control modes suitable for different flight conditions. This method upgrades the resource allocation from a "static mechanism" to a "dynamically adjustable" strategy without compromising attitude control quality. When RW angular momentum approaches the upper limit, aerodynamic surface unloading is triggered proactively, preventing irreversible failures common in traditional RW systems near saturation and ensuring continuity and stability of the composite execution chain.

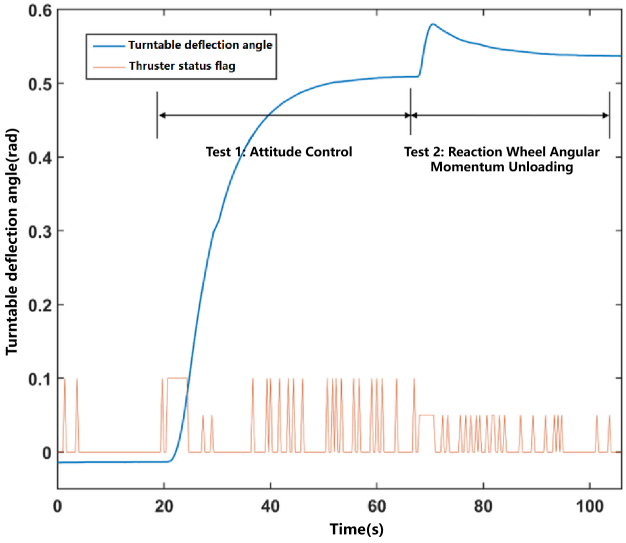


Fig. 15 Turntable deflection angle

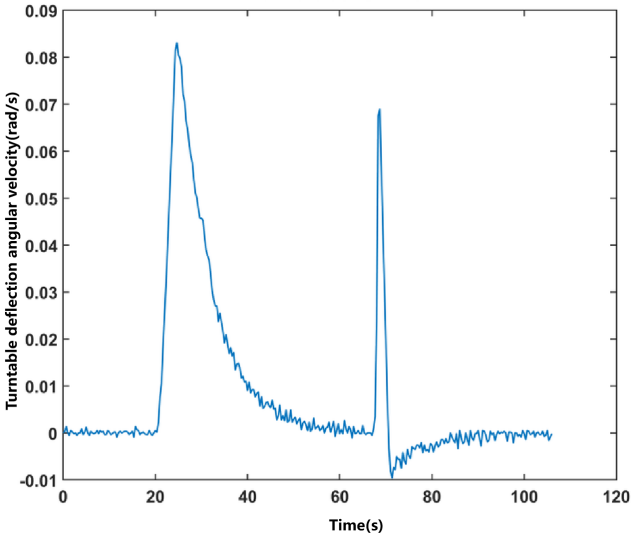


Fig. 16 Turntable angular velocity

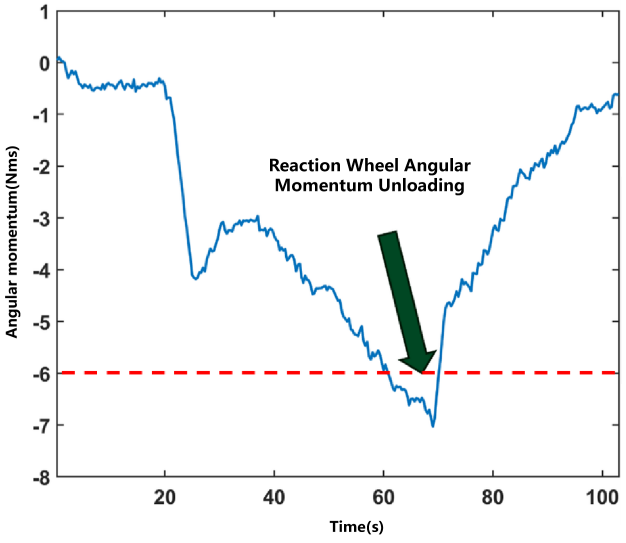


Fig. 17 Reaction wheel angular momentum

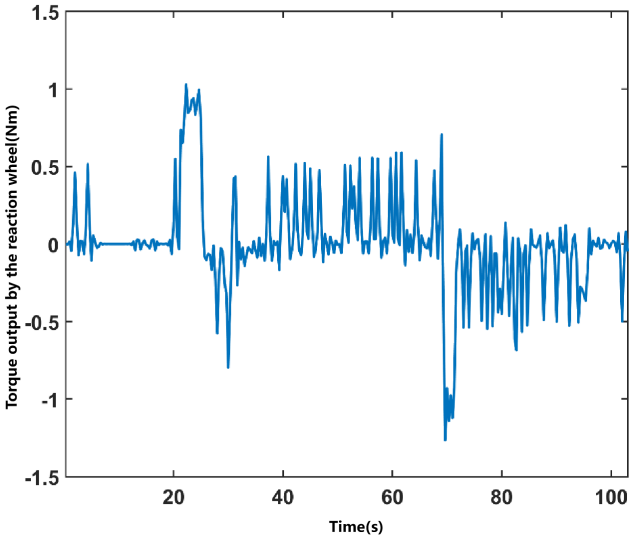


Fig. 18 Torque output by the reaction wheel

3. A reaction wheel angular momentum pre-biasing mechanism is introduced. During MPC prediction, if future aerodynamic surfaces are expected to be insufficient while current surfaces still have residual torque capacity, the RW can output negative torque in advance, "injecting" excess aerodynamic torque into its angular momentum reserve. This mechanism achieves cross-temporal energy storage and release, allowing the RW to provide larger transient control torques during surface authority degradation, thereby improving tracking accuracy and dynamic response.

In summary, the proposed composite attitude control theory and algorithm provide a novel solution for high-precision attitude control and rapid maneuvering in rarefied atmosphere environments. The results demonstrate that the "aerodynamic–reaction actuator resources" can be integrated, scheduled, and transferred across varying density conditions, offering valuable reference for future cross-domain atmospheric–near-space control research and engineering applications.

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